

Location of hydraulic resistance in the soil-plant pathway in seedlings of *Pinus sylvestris* L. grown in peat

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Seedlings of *Pinus sylvestris* L. were grown in three different soil media: 100% peat, 40% silt - 60% peat, and 60% silt - 40% peat. The percentages refer to total dry weight. Needle conductance, needle water potential, and plant water conductance were measured at different levels of soil water potentials controlled with a semipermeable membrane system. Seedlings grown in the 60:40 silt-peat mixture had a plant water conductance at a soil water potential of -0.1 MPa 3 times that of seedlings grown in pure peat. In an experiment where the roots were dipped in a silt slurry before planting, it was found that the plant water conductance at low soil water potential (-0.1 MPa) increased more than 2 times compared with undipped controls. We concluded that an important resistance to water flow in the soil-plant pathway was located in the soil outside the roots and probably was the most important resistance in the root-soil interface. The use of low humified peat as a growth medium is open to discussion from a silvicultural point of view because of its hydraulic properties under dry conditions.

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Des semis de *Pinus sylvestris* L. ont été produits dans trois substrats différents: mousse de sphagnum (100%), sable loameux (40%) et mousse de sphagnum (60%), sable loameux (60%) et mousse de sphagnum (40%). Les pourcentages se réfèrent ici au poids sec total. La conductance foliaire, le potentiel hydrique foliaire et la conductance hydrique ont été mesurés à différents niveaux de potentiels hydriques du sol. Les semis qui se sont développés dans le mélange sable loameux-mousse de sphagnum (60:40%) avaient une conductance hydrique trois fois supérieure à un potentiel hydrique du sol de $-0,1$ MPa comparativement à ceux qui se sont développés dans la mousse de sphagnum pure. Dans une expérience où les racines furent trempées dans une vase de sable loameux avant la plantation, il s'est avéré que la conductance hydrique de la plante, sous de faibles potentiels hydriques ($-0,1$ MPa), s'est amplifiée d'un facteur 2 comparativement aux semis témoins non trempés. Nous avons conclu qu'une importante résistance au débit de l'eau dans le continuum sol-plante est situé dans le sol à l'extérieur des racines représentant ainsi probablement la plus importante résistance dans l'interface racine-sol. A cause de ses propriétés hydrauliques sous des conditions de sécheresse, l'utilisation de la mousse de sphagnum faiblement humidifiée demeure un sujet ouvert pour discussion sur le plan silvicultural.

[Traduit par le journal]

Introduction

Seedlings grown in containers are widely used for reforestation in Sweden. The growing medium is usually low humified peat. However, earlier experiments with planted seedlings of *Pinus sylvestris* indicated an impaired water uptake in seedlings grown in peat (Örlander 1984; Örlander and Due 1986). The water uptake was also negatively affected by a small decrease in soil water potential (Örlander and Due 1986). A possible reason for this could be that water movement was restricted by low hydraulic conductivity in the peat or in the root-soil interface.

Difficulties in measurements of water flow and water potentials in the vicinity of the root have led to the construction of theoretical models to explain the hydraulic properties in the root and in the surrounding soil. The development of such models dates back to studies of Gardner (1960). Gardner's models have been improved and criticized by several authors (e.g., Cowan 1965; Newman 1969a, 1969b, 1974; Taylor and Klepper 1975). An excellent review concerning models of water transport in the soil-plant system is given by Molz (1981). The soil hydraulic conductivity, the rooting density, and the extraction rate are the most critical factors influencing the result. Some authors have found that the dominant resistance to water flow resides in the soil rather than in the root tissue (for references, see Molz 1981), while others (e.g., Newman 1969a, 1969b, 1974) put forward both theoretical and experimental evidence that the resistance in the root is the major one. It

is often found, however, that the total resistance in the soil-plant system increases with decreasing soil water potential (e.g., Faiz and Weatherly 1977, 1978; Herkelrath *et al.* 1977a, 1977b; Burch 1979), even though the theory implies the soil resistance to be negligible. Herkelrath *et al.* (1977b) therefore suggested the root-contact model for water flow to the root. This model assumes that the resistance to water flow is inversely proportional to the wetted fraction of the root surface. Consequently, if the soil pores are drained this would result in a considerable root-contact resistance. Root shrinkage would also give similar effects (cf. Huck *et al.* 1970).

In the present investigation an attempt has been made to locate the resistance in the hydraulic soil-plant pathway of seedlings grown in peat. The soil resistance was experimentally altered to test its influence on the total conductance in the hydraulic soil-plant pathway. To keep the climatic conditions and the soil water potentials constant, the experiments were performed in a growth chamber and a semipermeable membrane system (Örlander and Due 1986) was used to control soil water potentials and the soil temperature.

Materials and methods

The materials and methods used in the present experiments were similar to those used in some previous experiments (Örlander and Due 1986). Seedlings of Scots pine (*Pinus sylvestris* L.) were grown in containers (Paperpot 408, Lännen-Tehtaat OY, Finland) filled with three mixtures of low humified peat (Hasselfors super fine, Hasselfors, Sweden) and silt: (i) pure peat (ii) 40% silt - 60% peat, and (iii) 60%

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TABLE 1. Projected needle area (mean $\pm$ SE) and estimated root length of seedlings grown in various mixtures of peat and silt

Growing medium	Total needle area (cm ²)	Root length (cm/cm ³)
Peat	41.6 $\pm$ 3.2	13.6
40% silt	40.1 $\pm$ 2.0	12.1
60% silt	41.0 $\pm$ 3.8	13.6

silt–40% peat. The percentages refer to total dry weight. The seedlings were grown one growing season before the start of the experiment. The growing scheme resulted in similar needle area and root length of the seedlings irrespective of the growing medium (Table 1). The experiments were carried out in a growth chamber with a temperature of 20:10°C ($\pm$ 1°C) (day:night) and a photon flux density of ca 275 $\mu\text{mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$. Relative humidity was 50 $\pm$ 5% during the day and 90 $\pm$ 5% at night. The soil temperature was kept at ca. 15°C and the soil moisture was kept at predetermined levels by using a semipermeable membrane system (for details, see Örlander and Due 1986).

To acclimatize the seedlings to the climate in the growth chamber, they were planted in well-watered peat and kept for 5 days in the growth chamber. The soil temperature of the peat was approximately 25°C during the day. Thereafter the soil substrates of the seedlings were enclosed in semipermeable membranes and placed into six different concentrations of polyethylene glycol (PEG) 20 000: 0, 15, 30, 50, 70, and 100 g PEG/kg H₂O. Seven seedlings (expt. 2, six seedlings) were put in each of six vessels containing PEG and water. The water potential of the soil substrate will be somewhat lower than the potential of the solution when equilibrated (cf. Zur 1967). It was therefore necessary to water some of the seedlings immersed into pure water. This was made if soil water potentials decreased below –0.01 MPa.

Soil water potential (ψ_s) and soil temperature were measured inside the semipermeable membrane and recorded during the entire experimental period. The ψ_s was measured with miniature tensiometers except for the 70 and 100 g PEG solutions, where gypsum blocks were used. Needle conductance (g_n) and needle water potential (ψ_n) were measured by means of a null-balance porometer (Licor LI-1600, Lincoln, NE) and a Scholander pressure chamber (Djos, Uppsala, Sweden), respectively. All measurements were performed between 0800 and 1000, i.e., 5–7 h after the light was switched on. The projected needle area was measured with an areameter (Delta-T Ltd., Cambridge, U.K.). The length of the roots was estimated from their projected area, complemented with a measure of the mean root width. The plant water conductance (G_p) was calculated from the following equation (Hellkvist *et al.* 1980; cf. Whitehead and Jarvis 1981; Passioura 1982):

$$[1] \quad G_p = \frac{q}{\psi_s - \psi_n}$$

where q is the water flux per needle area, estimated from

$$[2] \quad q = g_n \text{VCD}$$

where VCD is the vapour concentration deficit.

Experimental design

Three experiments were designed to experimentally alter the soil resistance to test its influence on the hydraulic conductance in the soil–plant pathway. In expt. 1 the response to decreasing soil water potentials was evaluated for seedlings rooted in peat and two peat–silt mixtures. Expt. 2 was performed to examine whether the differences, if any, between the studied seedlings remained, despite the removal of the various growing media from the roots. The effect of increasing the conductivity at the root surface with a root coating of silt was tested in expt. 3.

TABLE 2. Initial values (means $\pm$ SE) of needle conductance (g_n), needle water potential (ψ_n), and plant water conductance (G_p)

Treatment	g_n (cm ² ·s ^{–1})	ψ_n (–MPa)	G_p ($\mu\text{g} \cdot \text{cm}^{-2} \cdot \text{s}^{-1} \cdot \text{MPa}^{-1}$)
Experiment 1, $n = 14$			
Peat	0.165 $\pm$ 0.011	0.81 $\pm$ 0.03	2.25 $\pm$ 0.20
40% silt	0.132 $\pm$ 0.018	0.86 $\pm$ 0.03	1.61 $\pm$ 0.24
60% silt	0.129 $\pm$ 0.012	0.87 $\pm$ 0.03	1.54 $\pm$ 0.15
Experiment 2, $n = 12$			
Peat	0.168 $\pm$ 0.015	0.79 $\pm$ 0.03	2.46 $\pm$ 0.17
40% silt	0.108 $\pm$ 0.007	0.77 $\pm$ 0.07	1.40 $\pm$ 0.11
60% silt	0.146 $\pm$ 0.010	0.87 $\pm$ 0.03	1.69 $\pm$ 0.13
Experiment 3, $n = 21$			
Controls	0.399 $\pm$ 0.015	0.70 $\pm$ 0.02	4.92 $\pm$ 0.25
Roots later dipped into silt	0.374 $\pm$ 0.014	0.72 $\pm$ 0.01	4.31 $\pm$ 0.18

NOTE: The seedlings were planted in well-watered peat ($\psi_s > -0.01$ MPa) and placed in the growth chamber. In expts. 2 and 3 the roots of the seedlings were washed before planting.

Experiment 1

After the removal of the paper walls, the seedlings were planted in well-watered peat. After 5 days in the growth chamber, the seedlings with intact substrates were inserted into the semipermeable membranes and put into the PEG solutions. The experiment included (i) seedlings grown in peat, (ii) seedlings grown in 60:40 peat–silt, and (iii) seedlings grown in 40:60 peat–silt. Seven days after the start of the experiment, the membrane was ruptured at the top of the soil pot and the PEG solution was allowed to penetrate the soil.

Experiment 2

The roots of the seedlings grown in different substrates (peat–silt mixtures) were carefully washed and planted in well-watered peat. After the 5-day acclimatization period, the seedlings were replanted in a new pot containing peat, inserted into semipermeable membranes, and placed in the PEG solutions.

Experiment 3

The roots of seedlings grown in pure peat were carefully washed and the seedlings were replanted in well-watered peat. After the 5-day acclimatization period, the roots of half of the seedlings were dipped into a slurry of water and silt. The roots were then placed in the middle of a new pot completely surrounded with peat. The other half of the seedlings were replanted and used as controls. The soil was allowed to dry according to the methods already described.

Initial value

On the 2 last days of the acclimatization period, g_n , ψ_n , and G_p were measured for all seedlings. The mean of the two values is the initial value (Table 2). In expts. 2 and 3 the initial values refer to seedlings with washed roots.

Statistical calculations

Mean values, standard deviations, and standard errors were calculated by using standard methods. Regression analysis was made by means of indicator variables. The following model was used:

$$[3] \quad \ln(y) = c + b_1 \cdot \ln(x) + b_2 \cdot D_1 \cdot \ln(x) + b_3 \cdot D_2 \cdot \ln(x)$$

D_1 and D_2 are indicator variables taking the values 0 or 1, depending on the treatment, and c , b_1 , b_2 , and b_3 are the regression coefficients. The difference between the coefficients was tested with statistical standard methods for linear models (Scheffé 1959). The equations were corrected for logarithmic bias.

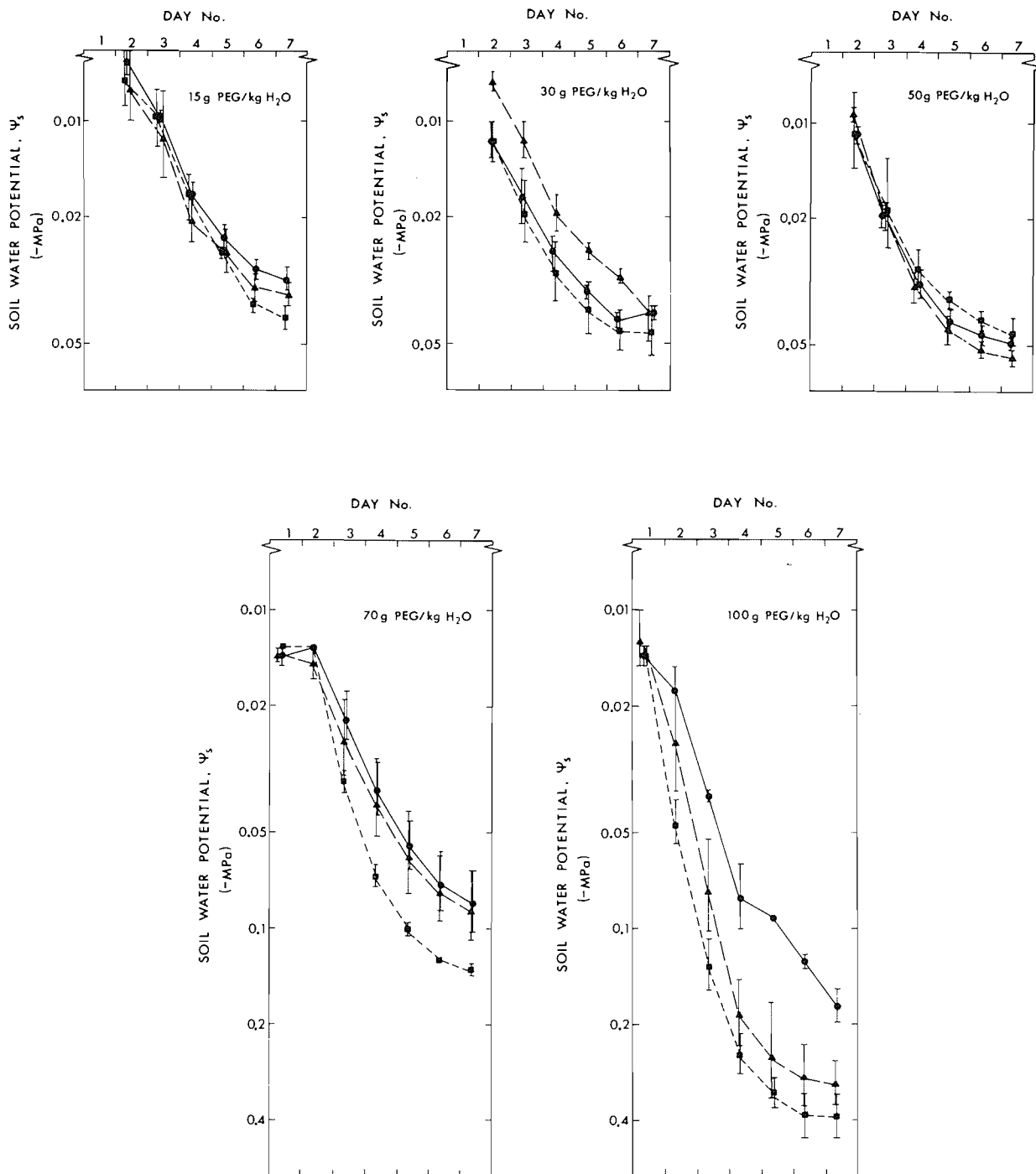


FIG. 1. Soil water potential (ψ_s) of Scots pine seedlings inserted into semipermeable membranes and placed in different concentrations of polyethylene glycol (PEG) on day 0 (expt. 1). The seedlings were rooted in different medias: ●, peat; ▲, 40:60 silt-peat; ■, 60:40 silt-peat. Vertical bars represent ± 1 SE. The ψ_s values are presented on a logarithmic scale.

Results

Experiment 1

The peat-silt mixtures often dried faster than the pure peat (Fig. 1). This resulted in lower soil water potentials, especially for the 60:40 peat-silt mixture in the 70 and 100 g PEG solutions. The g_n , ψ_n , G_p values varied only slightly between the treatments at the start of the experiment, when the seedlings were well watered (Table 2). The difference in G_p and g_n between the treatments increased with decreasing soil water

potentials, showing that seedlings grown in peat-silt mixtures had considerably higher G_p values at low soil water potentials (Fig. 2). Seedlings, especially those grown in 60% silt, maintained a relatively high water uptake even at low soil water potentials. At a soil water potential of -0.1 MPa, seedlings in 60% silt had a G_p that was 43% of the initial value, whereas the corresponding percentage for seedlings in pure peat was only about 14%. The differences between the slopes of the regression curves were significant ($p < 0.05$).

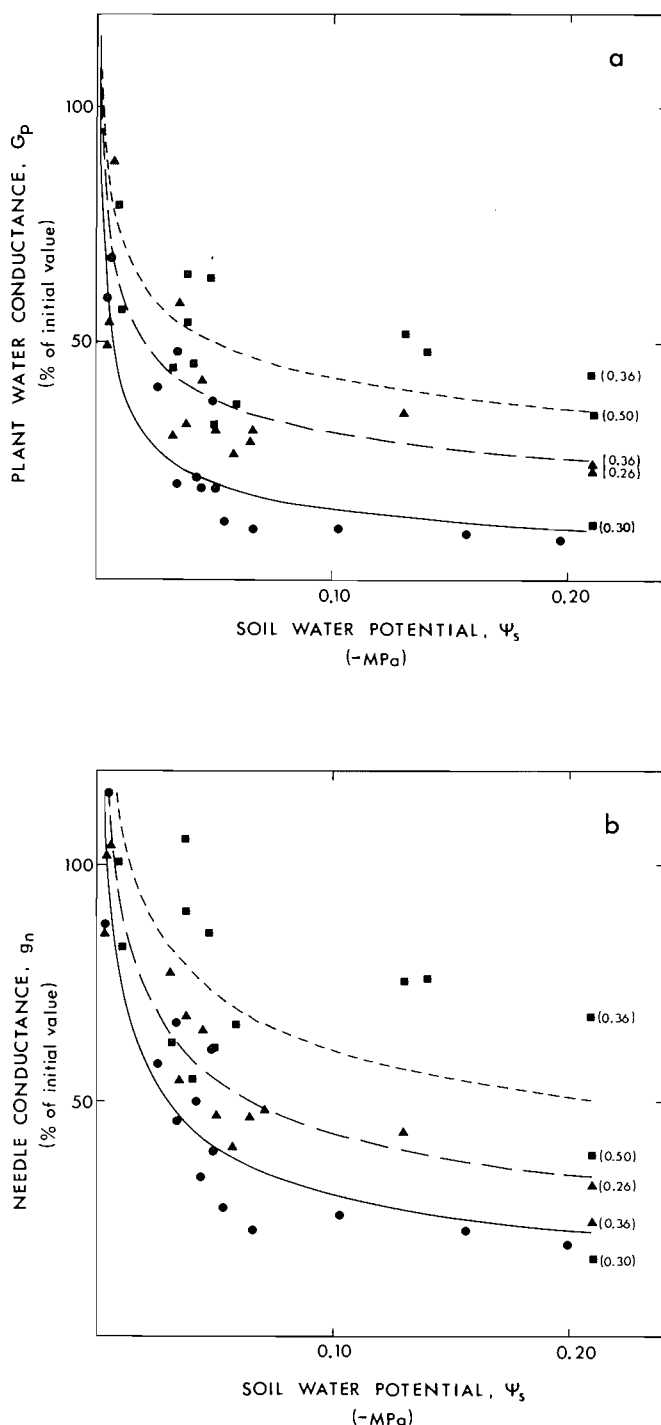


FIG. 2. The effect of soil water potential (Ψ_s) on (a) plant water conductance (G_p) and (b) needle conductance (g_n) of Scots pine seedlings rooted in different media (expt. 1). ●, peat; ▲, 40:60 silt-peat; ■, 60:40 silt-peat. Values were obtained on day 7 (cf. Fig. 1). Values of Ψ_s lower than -0.25 MPa are shown in parentheses.

The result of rupturing the semipermeable membrane is exemplified in Fig. 3. The G_p increased after the PEG solution had penetrated the soil, especially for seedlings grown in pure peat.

Experiment 2

The Ψ_s values were not fully stabilized at the end of expt. 2 (Fig. 4). However, there was only a slight decrease during the last 2 days. The soil water potentials were approximately equal,

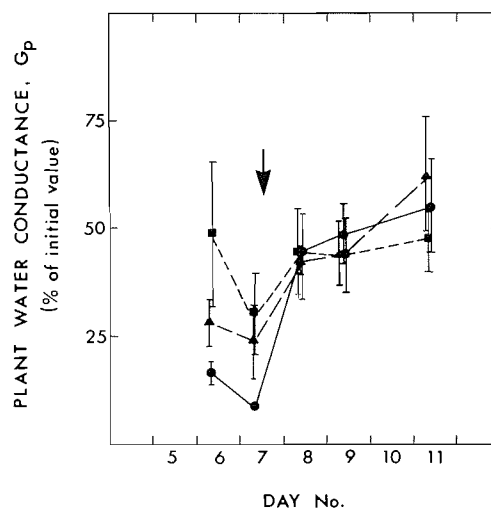


FIG. 3. Development of the plant water conductance (G_p) after rupturing the semipermeable membranes surrounding the soil pot of Scots pine seedlings placed in a solution of 100 g PEG/kg H_2O . The membranes were ruptured on day 7 (arrow). The osmotic potential of the solution was approximately -0.13 MPa (value derived from Williams and Shaykewich 1969). The seedlings were rooted in different media: ●, peat; ▲, 40:60 silt-peat; ■, 60:40 silt-peat. Vertical bars represent ± 1 SE.

irrespective of previous growing medium, except for seedlings in the 100 g PEG/kg H_2O solution, where seedlings previously grown in peat had higher mean water potentials compared with those previously grown in peat-silt mixtures (Fig. 4). G_p , g_n , and Ψ_s differed somewhat between the treatments at the start of the experiment, when the seedlings were well watered, g_n and G_p being higher in seedlings previously grown in peat (Table 2). G_p and g_n decreased greatly as the soil water potential decreased (Fig. 5). According to the regression curves, G_p was only 11-19% of the initial values at a soil water potential of -0.1 MPa. However, there were only small differences between the treatments.

Experiment 3

The soil dried considerably faster in the more concentrated PEG solutions when the roots had been dipped into the silt slurry (Fig. 6). The seedlings that were dipped into the silt slurry had a significantly ($p < 0.01$) higher G_p and g_n at low soil water potentials, despite there being only small differences between undipped and dipped seedlings when they were well watered (Fig. 7, Table 2). For example, at a soil water potential of -0.1 MPa, the G_p values of the regression curves were 15 and 37% of that at the start.

Discussion

By growing the seedlings in a peat-silt mixture, the hydraulic conductance in the soil-plant system was improved compared with that in pure peat at low soil water potentials (Fig. 2). The probable reason for the increase is that the conductivity in the soil surrounding the root or at the root-soil interface was improved. When the PEG solution was allowed to penetrate the soil, an increase in G_p was readily observed, especially in seedlings grown in peat (Fig. 3). This would also be expected if the resistance to water flow was located in the soil rather than in the root.

It could be argued that the difference in conductivity between the studied types of seedlings could be due to physiological-

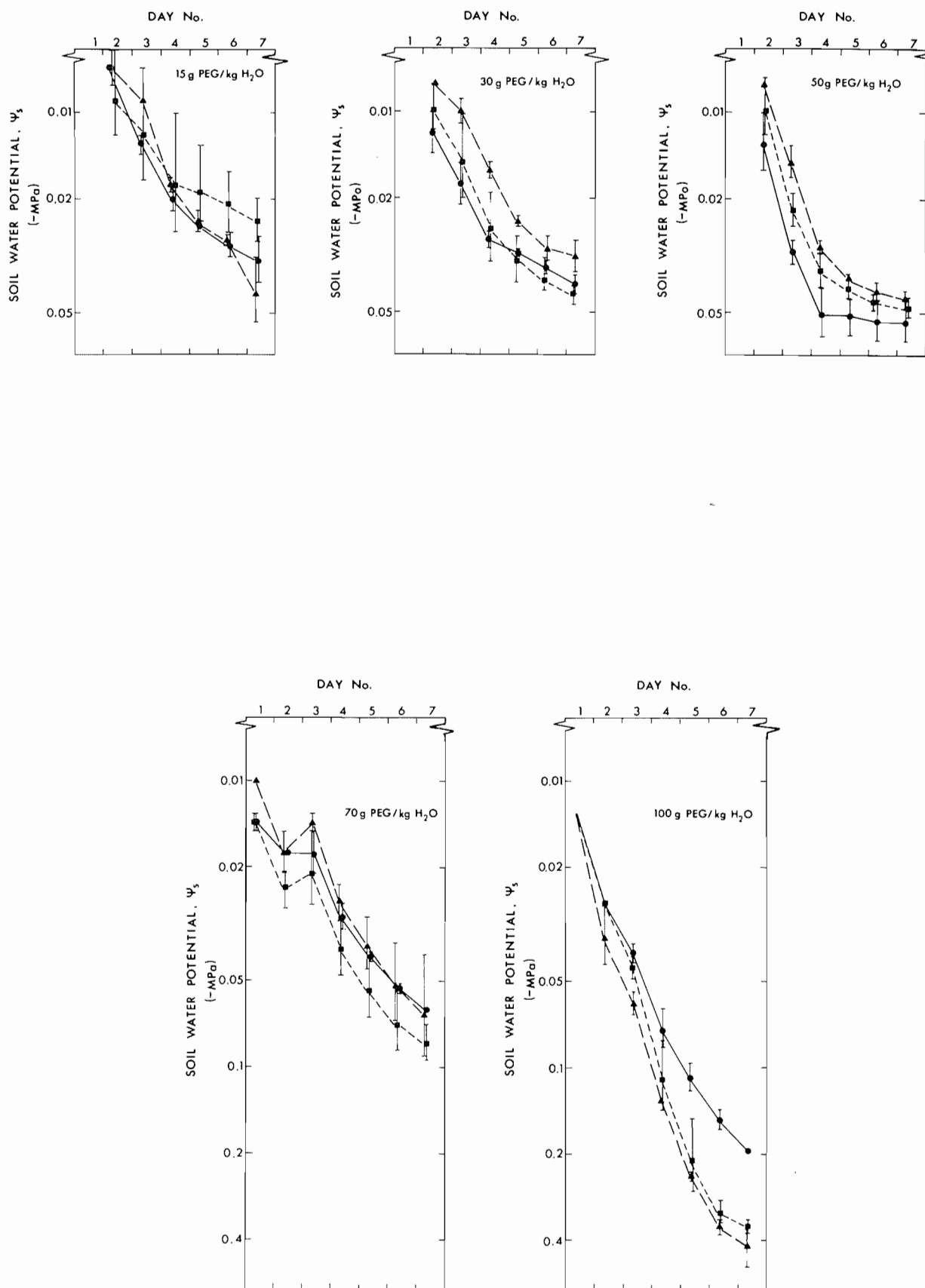


FIG. 4. Soil water potential (ψ_s) of Scots pine seedlings inserted into semipermeable membranes and placed in different concentrations of polyethylene glycol (PEG) on day 0 (expt. 2). The seedlings were grown in different soil media, but their roots were washed and planted in peat before being placed in the PEG solutions. ●, peat; ▲, 40:60 silt-peat; ■, 60:40 silt-peat. Vertical bars represent ± 1 SE. The ψ_s values are presented on a logarithmic scale.

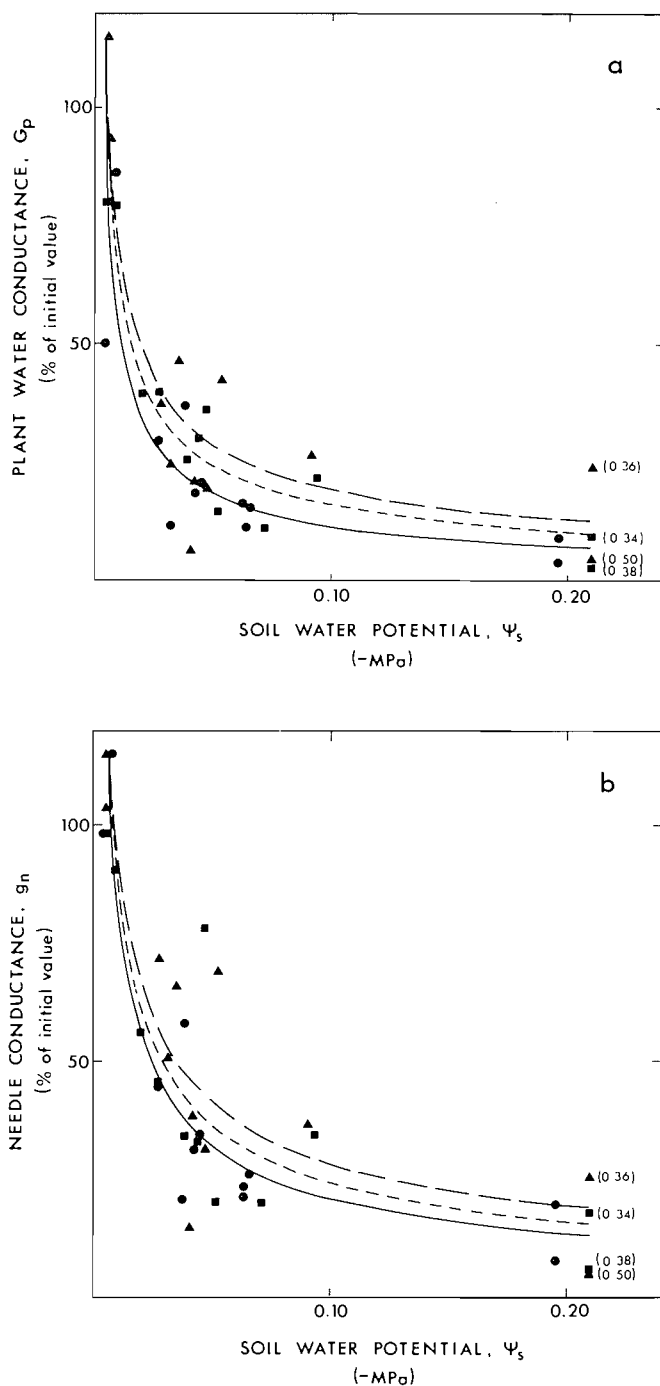


FIG. 5. The effect of soil water potential (ψ_s) on (a) plant water conductance (G_p) and (b) needle conductance (g_n) of Scots pine seedlings planted in peat but previously grown in different soil media (expt. 2). ●, peat; ▲, 40:60 silt-peat; ■, 60:40 silt-peat. Values were obtained on day 7 (cf. Fig. 4). Values of ψ_s lower than -0.25 MPa are shown in parentheses.

morphological differences. However, the seedlings had almost equal needle area and root length (Table 1), irrespective of growing medium. Furthermore, in expt. 2 (Fig. 5), when the soil had been washed away from the roots and all seedlings were planted in peat, the difference in conductivity between the seedlings was small. The low G_p and g_n for seedling grown in peat is therefore depending on the hydraulic characteristic of the

soil surrounding the roots and is not an indirect effect of the growing regime.

The results from expt. 3 indicate that the resistance was mainly in the root-soil interface, since the total conductance could be improved just by dipping the roots in silt slurry (Fig. 7).

The root-contact model, as presented by Herkelrath *et al.* (1977b), provides a good explanation of the results achieved in the present experiments. Since the pore size and thereby the water content at low ψ_s varies between soils of different textures, one would expect the contact resistance to vary with soil texture (cf. Dosskey and Ballard 1980; Bristow *et al.* 1984). Low humified peat has large pore spaces (cf. Päivänen 1973) and a mixture with silt will therefore result in smaller pores. Thus, compared with pure peat, the peat-silt mixture has more water-filled pores around the roots in dry soil resulting in a decreased contact resistance. An interesting fact is that experiments carried out in coarsely textured material, i.e., sand, often result in drastic decreases in transpiration as soil dries (e.g., Carbon 1973; Lüpke 1979; Zur *et al.* 1981). It is well known that peat shrinks when drying. Vapour gaps could thus be formed around the roots in dry peat and create an additional contact resistance. Furthermore, the roots shrink in dry soil (Huck *et al.* 1970), which might accentuate the problem still more.

The unsaturated conductivity in peat has not been widely studied. However, Bartels and Kuntze (1973) found that the conductivity decreased drastically with decreasing soil water potential and that the unsaturated conductivity in low humified peat was comparable with that in coarse sand. Similar results were obtained by Illner and Raasch (1977). The unsaturated conductivity in silt is considerably higher than in sand (cf. Marshall and Holmes 1979), probably several orders of magnitude higher, when the soil water potential decreases down below -0.1 MPa. The effect of mixing peat with silt is that the hydraulic conductivity will be much higher than in pure peat. It is thus possible to explain the differences in G_p at low ψ_s values obtained in expt. 1 (Figs. 2 and 3) in terms of different conductivity in the soil.

The root length and penetration of roots in the soil are important factors influencing the water uptake in the seedling. Most of the experiments cited above were carried out with herbs having higher root densities than woody plants (Newman 1969a). The soil conductivity would therefore theoretically restrict the water uptake more in woody plants, including *P. sylvestris*, than in other species when soil water potentials decrease, especially in newly planted seedlings.

From the present experiments it can be concluded that a considerable resistance in the soil-plant hydraulic pathway was located in the soil surrounding the roots and, in our opinion, the main resistance in the root-soil interface. The low water uptake generally found in planted seedlings (Örlander 1984) might be explained as a result of poor conductivity in the peat surrounding the roots or as a result of a contact resistance between root and soil. Growing media are chosen to give rapid growth with a minimum of growing problems in the nursery. The present investigation shows that growing media also have to be examined with respect to their functioning after planting. From a silvicultural standpoint, it can therefore be questioned whether peat or other coarsely textured soil media are suitable as growing medium for forest tree seedlings, except for soils at rather high matric potential.

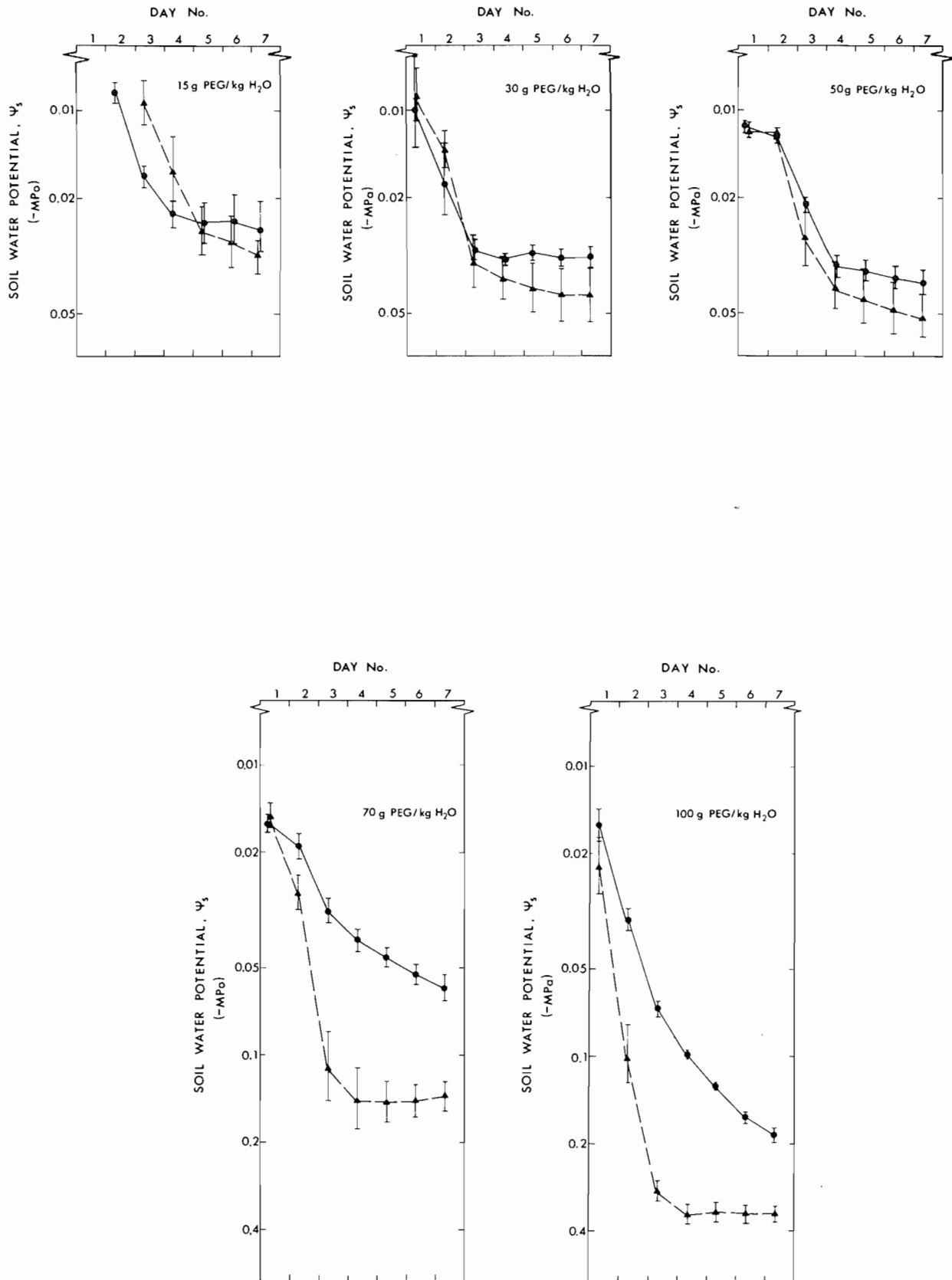


FIG. 6. Soil water potential (ψ_s) of Scots pine seedlings inserted into semipermeable membranes and placed in different concentrations of polyethylene glycol (PEG) on day 0 (expt. 3). The seedlings were grown in peat. Before being placed in the PEG solutions, the roots were washed, the roots of half of the seedlings were dipped into a slurry of silt and water, and all seedlings were finally replanted in peat. ●, control; ▲, roots dipped in silt. Vertical bars represent ± 1 SE. The ψ_s values are presented on a logarithmic scale.

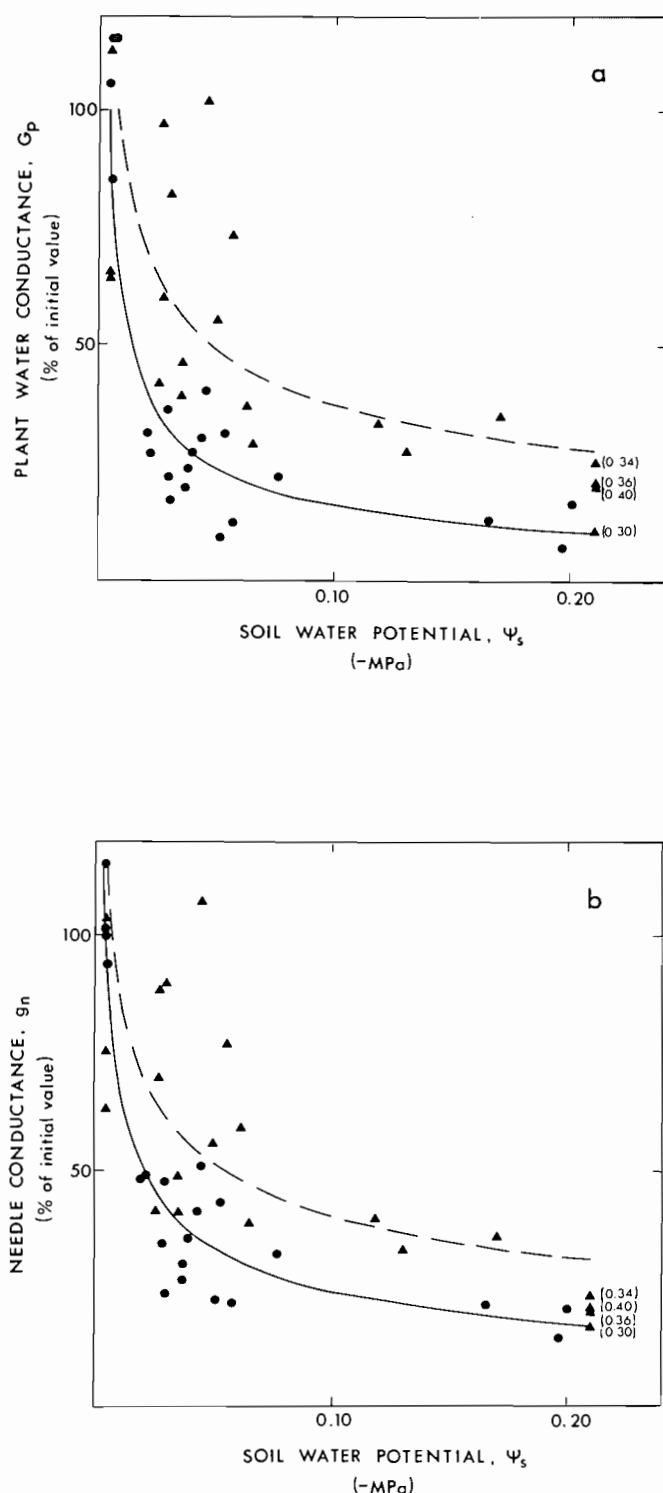


FIG. 7. The effect of soil water potential (Ψ_s) on (a) plant water conductance (G_p) and (b) needle conductance (g_n) of Scots pine seedlings planted in peat (expt. 3). The seedlings were previously grown in peat, their roots were washed, and the roots of half of the seedlings were dipped into a slurry of silt and water before being replanted. ●, control; ▲, roots dipped in silt. Values were obtained on day 7 (cf. Fig. 6). Values of Ψ_s lower than -0.25 MPa are shown in parentheses.

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- BARTELS, R., and H. KUNTZE. 1973. Torfeigenschaften und ungesättigte hydraulische Leitfähigkeit von Moorböden. *Z. Pflanzenernähr. Bodenkd.* **134**: 125–135.
- BRISTOW, K. L., G. S. CAMPBELL, and C. CALISENDORFF. 1984. The effects of texture on the resistance to water movement within the rhizosphere. *Soil Sci. Soc. Am. J.* **48**: 266–270.
- BURCH, G. J. 1979. Soil and plant resistances to water absorption by plant root systems. *Aust. J. Agric. Res.* **30**: 279–292.
- CARBON, B. A. 1973. Diurnal water stress in plants grown on a coarse soil. *Aust. J. Soil Res.* **24**: 33–42.
- COWAN, I. R. 1965. Transport of water in the soil–plant–atmosphere system. *J. Appl. Ecol.* **2**: 221–239.
- DOSSEY, M. G., and T. M. BALLARD. 1980. Resistance to water uptake by Douglas-fir seedlings in soils of different textures. *Can. J. For. Res.* **10**: 530–534.
- FAIZ, S. M. A., and P. E. WEATHERLY. 1977. The location of the resistance to water movement in the soil supplying the roots of transpiring plants. *New Phytol.* **78**: 337–347.
- . 1978. Further investigations into the location and magnitude of the hydraulic resistances in the soil–plant system. *New Phytol.* **81**: 19–28.
- GARDNER, W. R. 1960. Dynamic aspects of water availability to plants. *Soil Sci.* **89**: 63–73.
- HELLKVIST, J., K. HILLERDAL-HAGSTRÖMER, and E. MATTSOON-DJOS. 1980. Field studies of water relations on photosynthesis in Scots pine using manual techniques. *In* Structure and function of northern coniferous forests—an ecosystem study. *Edited by* T. Persson. *Ecol. Bull.* **32**: 183–204.
- HERKELRATH, W. N., E. E. MILLER, and W. R. GARDNER. 1977a. Water uptake by plants. I. Divided root experiments. *Soil Sci. Soc. Am. J.* **41**: 1033–1038.
- . 1977b. Water uptake by plants. II. The root contact model. *Soil Sci. Soc. Am. J.* **41**: 1039–1043.
- HUCK, M. G., B. KLEPPER, and H. M. TAYLOR. 1970. Diurnal variations in root diameter. *Plant Physiol.* **45**: 529–530.
- ILLNER, K., and H. RAASCH. 1977. Der Einfluss von Torfeigenschaften auf die kapillare Leitfähigkeit in Niedermoorböden. *Arch. Acker Pflanzenbau Bodenkd.* **10**: 753–758.
- LÜPKE, B. VON. 1979. Wasserhaushalt und Trockensubstanzproduktion bei Douglasien nach dem Verpflanzen. *Allg. Forst. Jagdztg.* **150**: 78–82.
- MARSHALL, T. J., and J. W. HOLMES. 1979. *Soil physics*. Cambridge University Press, Cambridge, U.K.
- MOLZ, F. J. 1981. Models of water transport in the soil–plant system: a Review. *Water Resour. Res.* **17**: 1245–1260.
- NEWMAN, E. I. 1969a. Resistance to water flow in soil and plant. I. Soil resistance in relation to amounts of root: theoretical estimates. *J. Appl. Ecol.* **6**: 1–12.
- . 1969b. Resistance to water flow in soil and plant. II. A review of experimental evidence on the rhizosphere resistance. *J. Appl. Ecol.* **6**: 261–272.
- . 1974. Root and soil water relations. *In* The plant root and its environment. *Edited by* E. W. Carson. University of Virginia, Charlottesville, VA. pp. 363–440.
- ÖRLANDER, G. 1984. Some aspects of water relations in planted seedlings of *Pinus sylvestris* L. Ph.D. dissertation, Swedish University of Agricultural Sciences, Umeå, Sweden.
- ÖRLANDER, G., and K. DUE. 1986. Water relations of Scots pine seedlings grown in peat as a function of soil water potential and soil temperature. *Stud. For. Succ.* In press.
- PÄIVÄNEN, J. 1973. Hydraulic conductivity and water retention in peat soil. *Acta For. Fenn.* **129**: 1–70.

- PASSIOURA, J. B. 1982. Water in the soil-plant-atmosphere continuum. *Encycl. Plant Physiol. New Ser.* **12B**: 5-33.
- SCHEFFÉ, H. 1959. *The analysis of variance*. John Wiley & Sons Ltd., London.
- TAYLOR, H. M., and B. KLEPPER. 1975. Water uptake by cotton root systems: an examination of assumptions in the single root model. *Soil Sci.* **120**: 57-67.
- WHITEHEAD, D., and P. G. JARVIS. 1981. Coniferous forests and plantations. *In* *Water deficits and plant growth*. Vol. 6. *Edited by* T. T. Kozłowski. Academic Press, New York.
- WILLIAMS, J., and C. F. SHAYKEWICH. 1969. An evaluation of polyethylene glycol (P.E.G.) 6000 and P.E.G. 20000 in the osmotic control of soil water matrix potential. *Can. J. Soil Sci.* **49**: 397-401.
- ZUR, B. 1967. Osmotic control of the matric soil-water potential II. Soil-plant system. *Soil Sci.* **103**: 30-38.
- ZUR, B., J. W. JONES, K. J. BOOTE, and L. C. HAMMOND. 1981. Total resistance to water flow in field soybeans. II. Limiting soil moisture. *Agron. J.* **74**: 99-105.